

THE PAPER 1: MEDICINE THROUGH TIME WITH THE WESTERN FRONT DOSSIER: MODERN

103 Crucial Questions. Complete Knowledge Mastery.

Operative / Student Name:

Target: Total Recall

RESTRICTED ACCESS - MASTER YOUR MEMORY

How to Hack Your Memory

(Rules of Engagement)

1. Never Just Read

Reading the answers feels like learning, but it's an illusion. You must force your brain to struggle to find the answer. The struggle is what builds memory pathways.

2. Cover, Recall, Check

Cover the answer sheet. Write your answer down in the notes section (or say it out loud). Only then are you allowed to check 'The Vault'.

3. Track Your Threat Level

Use the R-A-G (Red, Amber, Green) checkboxes next to every question after you attempt it.

-  **Green:** I nailed it instantly.
-  **Amber:** I got it, but I had to think hard.
-  **Red:** My mind went blank. (Revisit these tomorrow).

4. The Interrogation (Homework Protocol)

True mastery requires testing under pressure. Hand 'The Vault' (answers) to your parent or guardian. They will test you on 20 random questions. They hold the score; they hold the signature.

KT4.1: How have modern discoveries changed our understanding of illness?

Q#	The Target Question	R	A	G	Notes / My Answer
1	How did the 20th-century understanding of what causes illness fundamentally differ from the 19th-century focus on Germ Theory?	☐	☐	☐	
2	Which 19th-century scientist first demonstrated that physical characteristics could be passed down between generations of living things?	☐	☐	☐	
3	What crucial scientific technique did Rosalind Franklin and Maurice Wilkins use at King's College Hospital to help identify the structure of DNA?	☐	☐	☐	
4	In which year did James Watson and Francis Crick successfully work out and model the double-helix structure of DNA?	☐	☐	☐	
5	What major international scientific project was launched in 1990 to identify and map every single gene in human DNA?	☐	☐	☐	
6	What is a major modern limitation of the genetic breakthroughs achieved by mapping the human genome?	☐	☐	☐	
7	How do modern doctors use DNA discoveries to prevent or catch illnesses early through genetic screening?	☐	☐	☐	
8	According to modern scientific research, what specific medical condition can be triggered by eating a diet that is too high in sugar?	☐	☐	☐	
9	What chronic modern disease is directly linked to eating a diet	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
	too high in fat and experiencing obesity?				
10	Apart from lung cancer, what other debilitating respiratory disease has modern research linked directly to tobacco smoking?	☐	☐	☐	
11	Which chronic, organ-based illnesses has modern science linked directly to the excessive drinking of alcohol?	☐	☐	☐	
12	Which major piece of technology, invented in the 1930s, was 1,000 times more powerful than standard light microscopes, allowing scientists to see viruses for the first time?	☐	☐	☐	
13	How did diagnosis in the modern era fundamentally change compared to previous historical periods?	☐	☐	☐	
14	When were blood tests first introduced into modern medicine, and what was their diagnostic significance?	☐	☐	☐	
15	What is the key diagnostic advantage of a modern CT scan compared to a traditional X-ray?	☐	☐	☐	
16	When were ultrasound scans first developed, and how do they help doctors diagnose internal conditions like gallstones or kidney stones?	☐	☐	☐	
17	What are ECGs (electrocardiograms), which were developed in the 1900s, used for in modern diagnosis?	☐	☐	☐	
18	What is an endoscope (such as a bronchoscope used in lung investigations), and how does it assist in modern diagnosis?	☐	☐	☐	
19	What is a biopsy, and what role does it play in the modern	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
	diagnosis of chronic illnesses like cancer?				
20	When was blood sugar monitoring introduced, and how did it change the daily management of diabetes?	☐	☐	☐	

MISSION DEBRIEF

Score: _____ / 20

Parent/Guardian Authentication (Signature): _____

Date: _____

KT4.1: How have modern discoveries changed our understanding of illness?

Q#	The Target Question	R	A	G	Notes / My Answer
21	What disease did Edward Jenner find a vaccine for in 1796?	☐	☐	☐	
22	What did Jenner observe about milkmaids?	☐	☐	☐	
23	Who discovered the cause of cholera in 1854?	☐	☐	☐	
24	Explain how John Snow proved that cholera was a waterborne disease.	☐	☐	☐	
25	To what extent did the government support public health improvements in the mid-19th century?	☐	☐	☐	
26	In 1909, Paul Ehrlich and his research team developed Salvarsan 606 to treat syphilis. Why was Salvarsan 606 clinically known as the first 'magic bullet'?	☐	☐	☐	
27	During his research, how did Paul Ehrlich discover Salvarsan 606?	☐	☐	☐	
28	In 1932, Gerhard Domagk discovered Prontosil, the second 'magic bullet'. What type of infection did Prontosil cure?	☐	☐	☐	
29	How did 20th-century 'magic bullets' fundamentally differ in their clinical application from 19th-century antiseptics like Joseph Lister's carbolic acid?	☐	☐	☐	
30	What is the biological difference between early chemical 'magic bullets' like Salvarsan 606 and the first true antibiotic, penicillin?	☐	☐	☐	
31	After Gerhard Domagk published his findings on Prontosil, what active ingredient did French	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
	scientists isolate, leading to the creation of cheap sulphonamide drugs?				
32	Before the creation of the NHS, the Liberal Government introduced the 1911 National Insurance Act. What was a major limitation of this Act's medical coverage?	☐	☐	☐	
33	Which landmark 1942 government report recommended that the British state should provide comprehensive social security and healthcare 'from the cradle to the grave'?	☐	☐	☐	
34	In which year did the National Health Service (NHS) begin operating, and who was the government minister responsible for its launch?	☐	☐	☐	
35	What was the revolutionary founding principle of the National Health Service (NHS) in 1948?	☐	☐	☐	
36	When the NHS was launched in 1948, its services were organized into a 'tripartite' (three-part) system. What were these three parts?	☐	☐	☐	
37	Why did the British Medical Association (BMA)—representing the country's doctors—initially strongly oppose the creation of the NHS?	☐	☐	☐	
38	How did Aneurin Bevan famously win over the doctors who opposed the NHS?	☐	☐	☐	
39	How did the launch of the NHS in 1948 transform access to high-tech medical treatments for the working class?	☐	☐	☐	
40	Following the introduction of sulphonamide magic bullets in	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
	the 1930s and the NHS in 1948, what dramatic trend occurred in maternal mortality (mothers dying during or after childbirth)?				

MISSION DEBRIEF

Score: _____ / 20

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KT4.2: How have prevention and treatment advanced in the modern era?

Q#	The Target Question	R	A	G	Notes / My Answer
41	As part of its modern focus on preventing disease, what type of campaigns does the government run through the NHS?	☐	☐	☐	
42	Why does the 1935 illustration 'Medical moments in time' in the textbook emphasize that the pace of medical change in the first half of the 20th century was steady rather than rapid?	☐	☐	☐	
43	Which of the following is a successful example of a modern, government-funded NHS preventative immunization campaign that wiped out a major paralyzing disease in Britain?	☐	☐	☐	
44	Which of the following represents a modern, high-tech surgical treatment developed in the late 20th century that is provided free of charge under the NHS?	☐	☐	☐	
45	What is a major modern challenge facing the long-term effectiveness of antibiotic treatments like penicillin?	☐	☐	☐	
46	What structure did Watson and Crick discover in 1953?	☐	☐	☐	
47	What is the function of DNA in relation to disease?	☐	☐	☐	
48	Name one modern lifestyle factor that causes disease.	☐	☐	☐	
49	Explain how the discovery of DNA changed the understanding of disease.	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
50	Why did the mapping of the human genome (completed in 2003) not lead to immediate cures?	☐	☐	☐	
51	Recall: Who was Roger Bacon and what was their key contribution to medicine?	☐	☐	☐	
52	In which year did Alexander Fleming make his accidental observation of Penicillium mould on a staphylococci culture plate, and where did this take place?	☐	☐	☐	
53	What did Alexander Fleming actually observe in his forgotten petri dish in 1928 that led to the discovery of penicillin?	☐	☐	☐	
54	Why did Alexander Fleming abandon his research into penicillin as a systemic medical treatment by 1931?	☐	☐	☐	
55	Which of the following is a widespread historical 'myth' regarding the development of penicillin that examiners warn students to avoid in their essays?	☐	☐	☐	
56	How did Alexander Fleming's early laboratory experiments in 1928–1929 mislead him about penicillin's potential as an internal medicine?	☐	☐	☐	
57	Which two scientists revived Alexander Fleming's neglected 1929 paper on penicillin in 1938, and where were they based?	☐	☐	☐	
58	How did Ernst Chain contribute to the team's early breakthroughs in purifying penicillin?	☐	☐	☐	
59	What was the result of Florey and Chain's landmark scientific experiment on mice in the year 1940?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
60	Why did Florey and Chain's Oxford research team have to use household objects like milk churns, bed pans, and even a bath tub in their laboratory?	☐	☐	☐	

MISSION DEBRIEF

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KT4.3: How was Penicillin discovered and mass-produced?

Q#	The Target Question	R	A	G	Notes / My Answer
61	Who was the first human patient to be treated with Florey and Chain's purified penicillin in 1941, and how did he contract his infection?	☐	☐	☐	
62	What tragic clinical outcome occurred during the first human trial of penicillin in 1941, and why?	☐	☐	☐	
63	When Florey and Chain first approached British pharmaceutical companies in 1939–1940 to mass-produce penicillin, why were they turned down?	☐	☐	☐	
64	To which country did Howard Florey travel in July 1941 to seek the industrial help and investment needed to mass-produce penicillin?	☐	☐	☐	
65	What major historical event in December 1941 served as a catalyst, prompting the US government to heavily fund the mass production of penicillin?	☐	☐	☐	
66	Which industrial manufacturing method allowed US pharmaceutical companies to scale up penicillin production from small laboratory bottles to millions of doses?	☐	☐	☐	
67	How many doses of penicillin did Allied medical services have available by the D-Day landings in June 1944, demonstrating the success of wartime mass-production?	☐	☐	☐	
68	How did Howard Florey's attitude toward patenting penicillin accelerate its worldwide availability and keep manufacturing costs down?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
69	Which female scientist at Oxford University identified the exact chemical structure of penicillin in 1945, allowing chemists to create semi-synthetic variations?	☐	☐	☐	
70	In which year did Fleming, Florey, and Chain jointly receive the Nobel Prize in Medicine to recognize their collective roles in the penicillin breakthrough?	☐	☐	☐	
71	Looking at the penicillin story through Edexcel GCSE 'factors', which combination of factors was most crucial in turning a minor 1928 laboratory discovery into a global medical revolution by 1944?	☐	☐	☐	
72	What is a 'magic bullet'?	☐	☐	☐	
73	Name the first magic bullet discovered by Paul Ehrlich in 1909.	☐	☐	☐	
74	What was established in Britain in 1948 to improve public health?	☐	☐	☐	
75	Explain the impact of the NHS on medical treatment in Britain.	☐	☐	☐	
76	How did technology improve diagnosis in the 20th century?	☐	☐	☐	
77	Recall: Who was Ernst Chain and what was their key contribution to medicine?	☐	☐	☐	
78	In which year did the British Medical Research Council publish the landmark study by Doll and Hill that conclusively proved the link between smoking and lung cancer?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
79	Why was the British government initially very slow to take legislative action against smoking after the scientific link to lung cancer was proven in 1950?	☐	☐	☐	
80	What was a major limitation of using traditional chest X-rays to diagnose lung cancer before advanced scanning technology was developed?	☐	☐	☐	

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Parent/Guardian Authentication (Signature): _____

Date: _____

KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

Q#	The Target Question	R	A	G	Notes / My Answer
81	What key technological advantage does a modern CT (computerised tomography) scan have over a traditional chest X-ray in diagnosing lung cancer?	☐	☐	☐	
82	How does a high-tech PET (positron emission tomography) scan assist modern doctors in managing a lung cancer patient's treatment?	☐	☐	☐	
83	What is bronchoscopy (a type of endoscope) and how is it used to diagnose lung cancer?	☐	☐	☐	
84	What is a biopsy, and why is it a crucial step in the modern diagnosis of lung cancer?	☐	☐	☐	
85	How does modern chemotherapy treat lung cancer, and what is its main clinical drawback?	☐	☐	☐	
86	What does modern radiotherapy treatment for lung cancer involve?	☐	☐	☐	
87	What is immunotherapy, one of the most advanced treatments for lung cancer in the twenty-first century?	☐	☐	☐	
88	What was the very first major advertising restriction the British government placed on tobacco, introduced in 1965?	☐	☐	☐	
89	Following gradual extensions of advertising rules, in which year did the government finally implement a complete ban on all tobacco advertising and sports sponsorship in the UK?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
90	In 2007, what legislative measure did the government take to restrict teenagers' access to tobacco products?	☐	☐	☐	
91	Under the Health Act of 2006, what major legislative ban came into force on 1 July 2007 to protect the public from passive smoking?	☐	☐	☐	
92	What retail display regulation was introduced in 2012 to discourage young people from taking up smoking?	☐	☐	☐	
93	In 2015, how did the government extend the smoking ban to protect children from the dangers of second-hand smoke?	☐	☐	☐	
94	Which of the following was a 'nudge' tactic introduced in 2015 that standardizes cigarette boxes to make them unappealing to young people?	☐	☐	☐	
95	Why is there currently no national routine screening program for lung cancer in the UK, unlike other chronic conditions?	☐	☐	☐	
96	When comparing government action against 19th-century cholera with 20th-century lung cancer, what major similarity do historians identify?	☐	☐	☐	
97	The government's modern campaigns (like the Sugar Tax and Change4Life) and legislative bans represent a transition away from which historical political philosophy?	☐	☐	☐	
98	Who discovered Penicillin by accident in 1928?	☐	☐	☐	
99	Which two scientists developed Penicillin into a usable drug?	☐	☐	☐	

Q#	The Target Question	R	A	G	Notes / My Answer
100	What role did the US government play in the mass production of Penicillin?	☐	☐	☐	

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Date: _____

KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

Q#	The Target Question	R	A	G	Notes / My Answer
101	Explain why Penicillin is considered a turning point in medicine.	☐	☐	☐	
102	To what extent was Fleming solely responsible for the success of Penicillin?	☐	☐	☐	
103	Recall: Who was Roger Bacon and what was their key contribution to medicine?	☐	☐	☐	

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THE VAULT

Restricted Access: Answer Keys

KT4.1: How have modern discoveries changed our understanding of illness?

1. It moved to a multi-causal approach, recognizing that while microbes cause infectious diseases, genetics and lifestyle choices cause chronic illnesses.
2. Gregor Mendel
3. X-ray crystallography to take photographs of human cells
4. 1953
5. The Human Genome Project
6. There is still no cure or highly effective treatment for the majority of genetic conditions.
7. By checking an individual's genetic code to see if they carry genes that put them at higher risk of developing certain cancers or conditions.
8. Type 2 diabetes
9. Heart disease
10. Emphysema
11. Liver disease, kidney disease, and several cancers
12. The Electron microscope
13. It shifted from a doctor's clinical observation of physical symptoms to precise, high-tech laboratory tests and scans.
14. The 1930s; they allowed doctors to test for a vast number of conditions without invasive surgery.
15. A CT scan is a 3D scan that creates detailed images of the inside of the body, allowing doctors to spot tiny tumors when they are the size of a pea.
16. The 1940s; they use high-frequency sound waves to build up a picture of the inside of the body.
17. They use electrical impulses to track and monitor heart activity.
18. A flexible tube with a camera that allows doctors to see inside the body and collect cell samples without major surgery.
19. Taking a tiny sample of tissue from a lump to test it and see if it is cancerous.
20. The 1960s; it allowed patients to regularly check their own blood sugar at home to manage their condition.
21. Smallpox.
22. He noticed that milkmaids who had caught cowpox never caught the deadly smallpox.
23. John Snow.

24. He mapped out cholera deaths in Soho and traced them all to a single infected water pump on Broad Street, proving it wasn't caused by miasma.
25. Initially, they maintained a 'laissez-faire' (hands-off) attitude, but the Great Stink (1858) and Snow's findings eventually forced them to act, such as building the London sewer system.
26. Because it was a synthetic chemical compound designed to target and destroy a specific disease-causing microbe inside the body without harming the rest of the patient's cells.
27. By methodically testing over 600 arsenic chemical compounds until finding one (the 606th) that specifically destroyed syphilis bacteria.
28. Puerperal (childbed) fever and blood poisoning caused by streptococcus bacteria.
29. Antiseptics killed microbes on external surfaces or open wound tissue, whereas magic bullets were taken internally to target and destroy specific pathogens inside the body without poisoning the patient.
30. Magic bullets were synthetic chemical compounds created in laboratories, whereas antibiotics like penicillin were created using living microorganisms to kill bacteria.
31. Sulphanilamide, which allowed pharmaceutical companies to mass-produce cheap, effective drugs for pneumonia, meningitis, and scarlet fever.
32. It only provided free GP care to sick workers who paid weekly contributions, completely excluding their wives, children, and those who were unemployed or retired.
33. The Beveridge Report
34. 1948; under Aneurin Bevan
35. That comprehensive healthcare should be free at the point of delivery for everyone in Britain, paid for through national taxation.
36. Primary care (GPs, dentists, opticians), hospital services (managed by regional boards), and local authority health services (such as ambulances and health visitors).
37. They feared they would lose their independent status, lose their private patient income, and become state-controlled civil servants.
38. By 'stuffing their mouths with gold'—allowing hospital consultants to continue treating lucrative private patients alongside their NHS work.
39. It made advanced diagnostic tests and specialized surgical treatments free for everyone, removing the fear of bankruptcy from getting ill.
40. It fell rapidly, dropping from high rates in the early 20th century to less than 1% by the late 20th century.

KT4.2: How have prevention and treatment advanced in the modern era?

41. Public education and lifestyle campaigns encouraging healthy eating, physical exercise, and reducing tobacco or alcohol consumption.
42. Because while breakthroughs like X-rays and aseptic surgery existed, many traditional home remedies (like wrapping brown paper and vinegar for headaches) remained highly common.
43. The introduction of the polio vaccine in the 1950s.
44. Organ transplants, prosthetic joint replacements, and keyhole surgery.
45. The evolution of drug-resistant strains of bacteria (superbugs like MRSA) due to the over-prescription and overuse of antibiotics.
46. The double helix structure of DNA.
47. DNA carries genetic information; mutations or inherited genetic defects can cause hereditary diseases.
48. Smoking, poor diet (obesity), or lack of exercise.
49. It allowed scientists to finally understand conditions that were neither infectious nor caused by lifestyle, such as Down's Syndrome or Cystic Fibrosis.
50. Identifying the genes responsible for a disease is only the first step; creating gene therapies to fix or replace them is vastly more complex.
51. Roger Bacon was a 13th-Century Scientist. An English philosopher and Franciscan friar who emphasized the study of nature through empirical observation. He was imprisoned by Church leaders for suggesting that doctors should conduct their own independent experiments.
52. 1928; at St Mary's Hospital in London
53. A contaminating mould had grown on the dish and killed off the surrounding staphylococci bacteria.
54. He lacked the specialized chemical team and funding needed to isolate, concentrate, and stabilize the active compound.
55. That Alexander Fleming single-handedly developed penicillin into a mass-produced, usable medicine.
56. He found that penicillin became completely ineffective when mixed with blood in test tubes, causing him to doubt its value in living people.
57. Howard Florey and Ernst Chain; at Oxford University
58. As a skilled biochemist, he successfully grew the mould in his laboratory and extracted the active chemical compound.
59. It proved that penicillin was a highly effective antibiotic capable of killing deadly infections inside living bodies.
60. Penicillin was incredibly difficult to produce in large quantities, requiring massive volumes of grown mould to extract tiny amounts of active liquid.

KT4.3: How was Penicillin discovered and mass-produced?

61. Albert Alexander, a local policeman who developed septicaemia after being scratched by a rose bush.

62. The patient initially showed dramatic recovery, but died after the tiny supply of penicillin ran out, despite the doctors trying to recycle the drug from his urine.
63. British firms were fully occupied with the war effort, manufacturing essential military supplies and ammunition for World War II.
64. The United States
65. The entry of the United States into World War II, making the treatment of wounded soldiers a matter of urgent national priority.
66. Growing the mould in giant beer vats and deep-fermentation tanks.
67. Over 2.3 million doses, enough to treat all Allied casualties
68. He refused to patent the drug, declaring that it should be a free, lifesaving resource available to everyone in the world.
69. Dorothy Crowfoot Hodgkin
70. 1945
71. Individual brilliance (Fleming/Florey/Chain), institutions (the US Government), science/technology (deep fermentation), and the catalyst of war (WWII).
72. A chemical compound that targets and kills specific disease-causing microbes in the body without harming the rest of the patient.
73. Salvarsan 606 (used to treat syphilis).
74. The National Health Service (NHS).
75. It made healthcare free at the point of delivery, dramatically improving access to treatments and hospitals for the working classes.
76. Technologies like X-rays, MRI scans, and blood tests allowed doctors to diagnose internal issues accurately without relying solely on patient symptoms or exploratory surgery.
77. Ernst Chain was a Biochemist. A brilliant German-Jewish refugee who worked alongside Howard Florey at Oxford. Chain successfully discovered the complex chemical method required to extract and purify Penicillin from the mould so it could be safely tested on mice and humans.
78. 1950
79. Because the government earned around £4 billion from tobacco tax, thousands of jobs depended on the industry, and there were ethical debates about limiting personal freedoms.
80. They were not clear enough, meaning tumors were often missed entirely or mistaken for other things like lung abscesses.

KT4.4: How has the government tackled the modern epidemic of Lung Cancer?

81. A CT scan uses dye and takes detailed 3D images, allowing doctors to spot tiny tumors when they are the size of a pea.
82. It shows how active the cancer cells are by tracking how much energy the tumor is using, helping doctors see if a treatment is working.
83. A flexible tube with a camera passed down the patient's airway, allowing doctors to see inside the lungs and collect cell samples.
84. It is the extraction and scientific analysis of a tiny sample of lung tissue to confirm if a tumor is indeed cancerous.
85. It uses powerful drugs (chemicals) to attack and kill cancer cells throughout the body, but it also damages healthy cells, causing severe side effects.
86. Using targeted, high-energy radiation beams to shrink and destroy localized cancer tumors.
87. Using specialized drugs to 'wake up' and boost the patient's own immune system so it can identify and destroy cancer cells.
88. Banning cigarette advertisements on television.
89. 2005
90. They raised the legal age for buying tobacco from 16 to 18.
91. A ban on smoking in all enclosed workplaces, including pubs, cafés, restaurants, and offices.
92. A requirement that all tobacco products in shops must be removed from open display and kept hidden from sight.
93. By banning smoking in all cars carrying children under the age of 18.
94. Compulsory plain, olive-green packaging featuring prominent graphic health warnings and no brand logos.
95. Because the available tests are not accurate enough to outweigh the negative effects of screening, such as exposing healthy people to radiation.
96. In both cases, the initial response was slow, with the government only directly intervening with compulsory laws after the death toll became too high to ignore.
97. The laissez-faire (leave it alone) approach of the 19th century
98. Alexander Fleming.
99. Howard Florey and Ernst Chain.
100. They funded the mass production of Penicillin so it could be used to treat Allied soldiers during World War II.
101. It was the first true antibiotic, capable of curing a wide range of bacterial infections that were previously fatal, saving millions of lives.
102. Fleming made the initial observation, but he failed to develop it. Florey, Chain, and US funding were essential to turning it into a life-saving drug.
103. Roger Bacon was a 13th-Century Scientist. An English philosopher and Franciscan friar who emphasized the study of nature through empirical observation. He was imprisoned by Church leaders for suggesting that doctors should conduct their own independent experiments.

Mastery Tracker

Track your total recall score across all 103 questions.

Attempt	Date	Score (/103)	Parent/Guardian Signature	Target for Next Time
1				
2				
3				

Operative Reflection

My ultimate strength in this unit is...

The three specific facts I need to hunt down and memorize tonight are...